

VISUAL REASONING AND MENTAL MAPS

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/ DEV @ DIGITAL NATIVES
/ LEAD DEV @ FLYIIN

MOTIVATION

DEFINITIONS

VISUAL REASONING

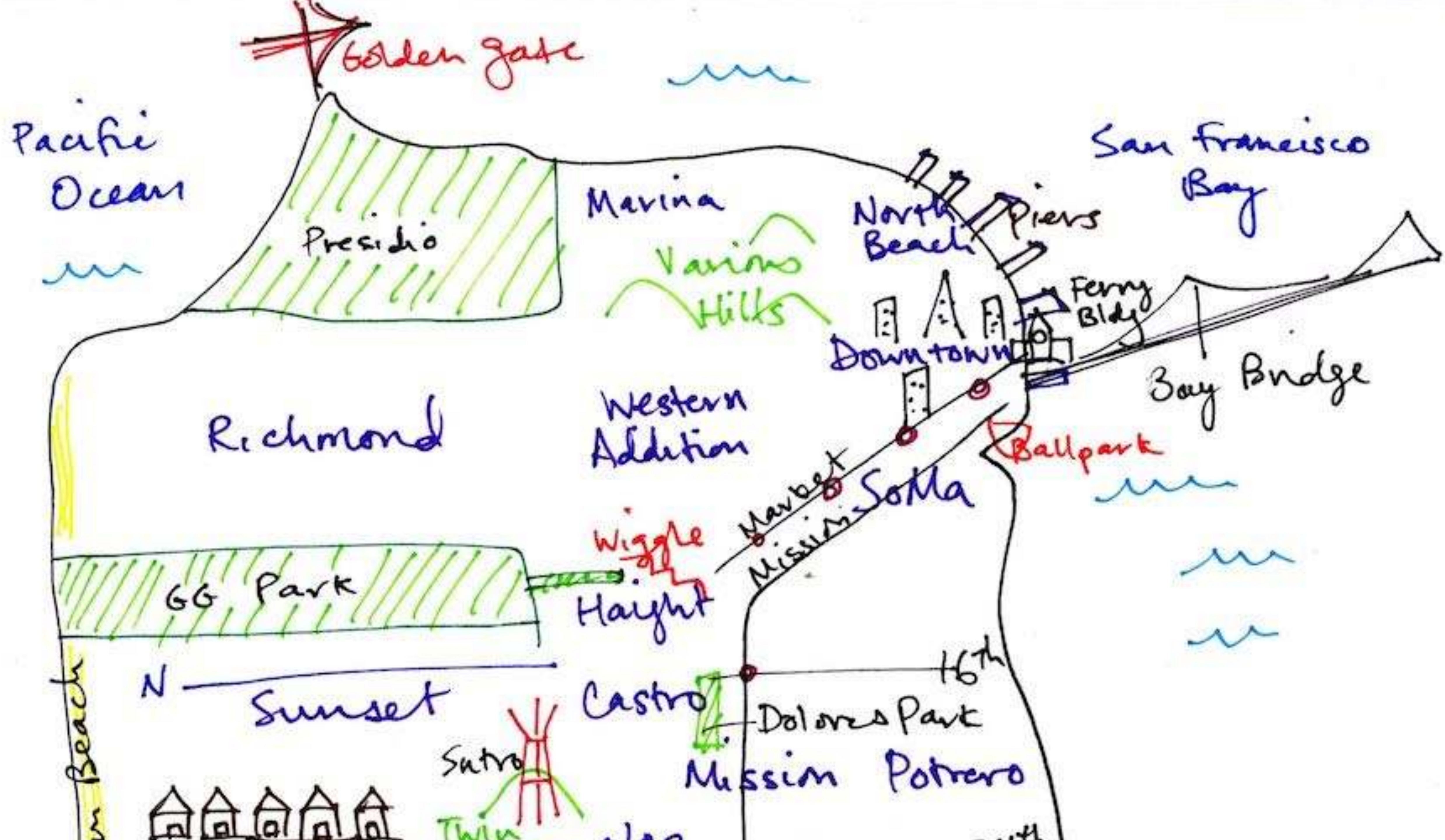
THE ACT OF COLLECTING INTUITIVE

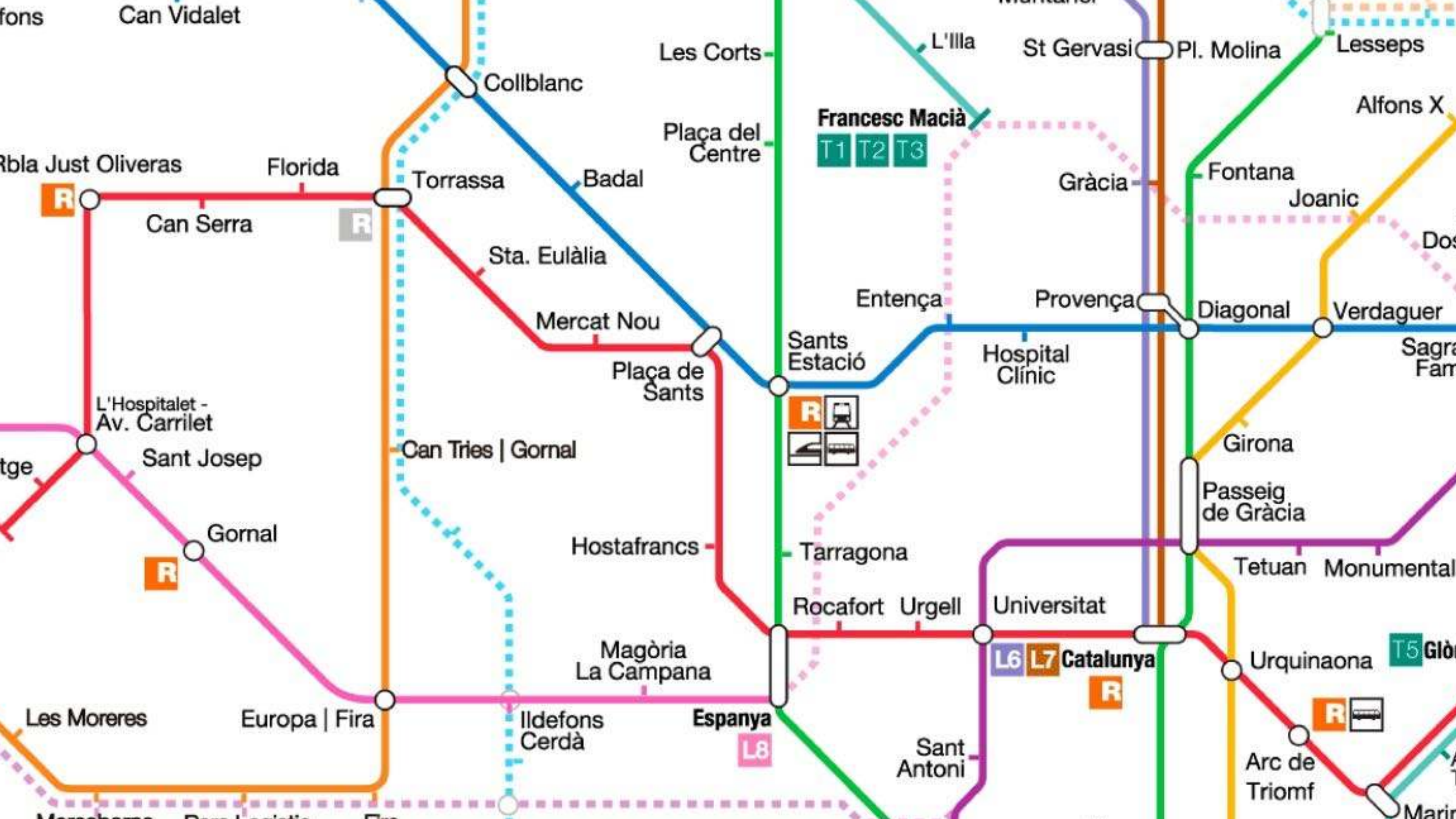
INFORMATION

AND GET TO A CONCLUSION

MENTAL MAPS

HOW OUR MIND DIGEST
THE COMPLEXITY OF SPATIAL
INFORMATION





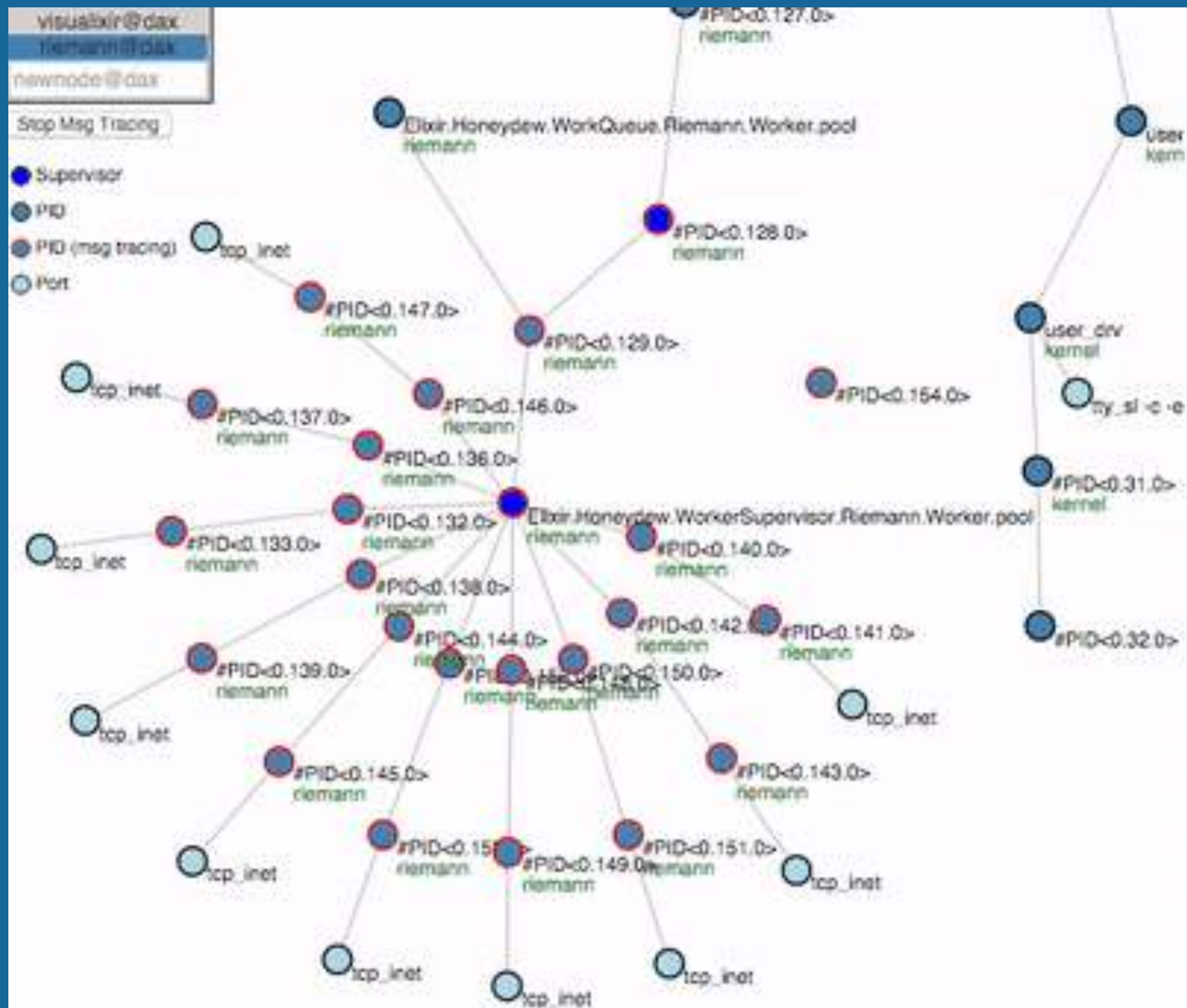
**A VISUAL TURN?
AGAIN?
THE LAST ONE?**

COMPETITIVE ANALYSIS OF VISUAL TOOLS FOR ERLANG/ELIXIR

VISUALIXIR

github.com/koudelka/visualixir

A process/message visualizer for BEAM nodes.



#PID<0.596.0> spawned

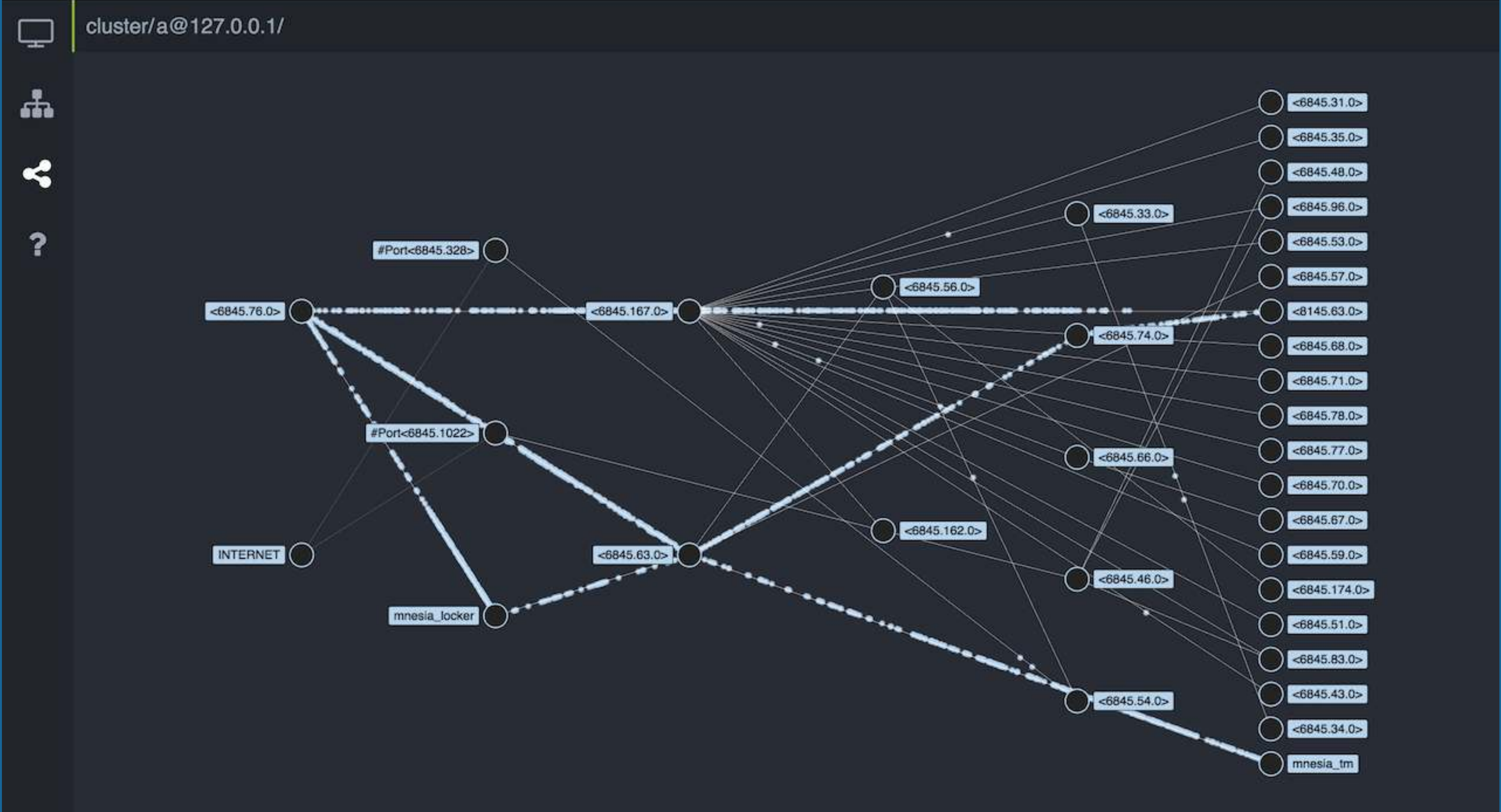
#PID<0.154.0> sent #PID<0.32.0> the message {revalied, #PID<15362.154.0>,
%EX.State{binding: [], cache: [], counter: 126, env: %Macro.Env{aliases: [], context:
nil, context_modules: [], export_vars: nil, files: "lex", function: nil, functions:
[{Kernel, [:=: 2, !=: 2, *: 2, +: 1, +: 2, ++: 2, -: 1, -: 2, --: 2, /: 2, <: 2, <=:
2, ==: 2, ===: 2, ==: 2, >: 2, >=: 2, abs: 1, apply: 2, apply: 3, binary_part: 3,
bit_size: 1, byte_size: 1, div: 2, elem: 2, exit: 1, function_exported?: 3,
get_and_update_in: 3, get_in: 2, hd: 1, inspect: 1, inspect: 2, is_atom: 1, is_binary:

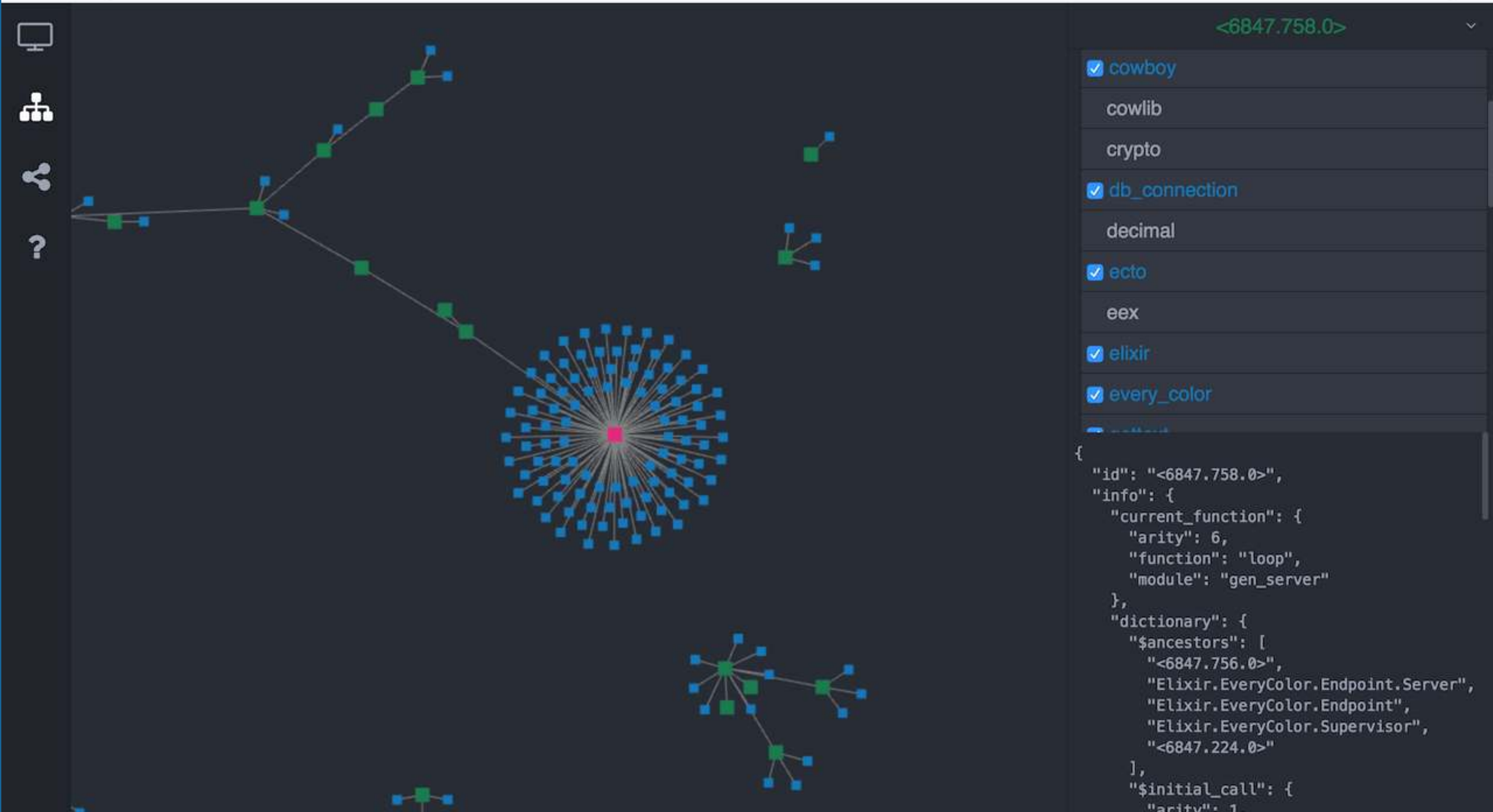
ERLANG PERFORMANCE LAB

<http://www.erlang.pl>

Apply Netflix's intuition engineering principles to the BEAM







<6847.758.0>

☒ cowboy

cowlib

crypto

☒ db_connection

decimal

☒ ecto

eex

☒ elixir

☒ every_color

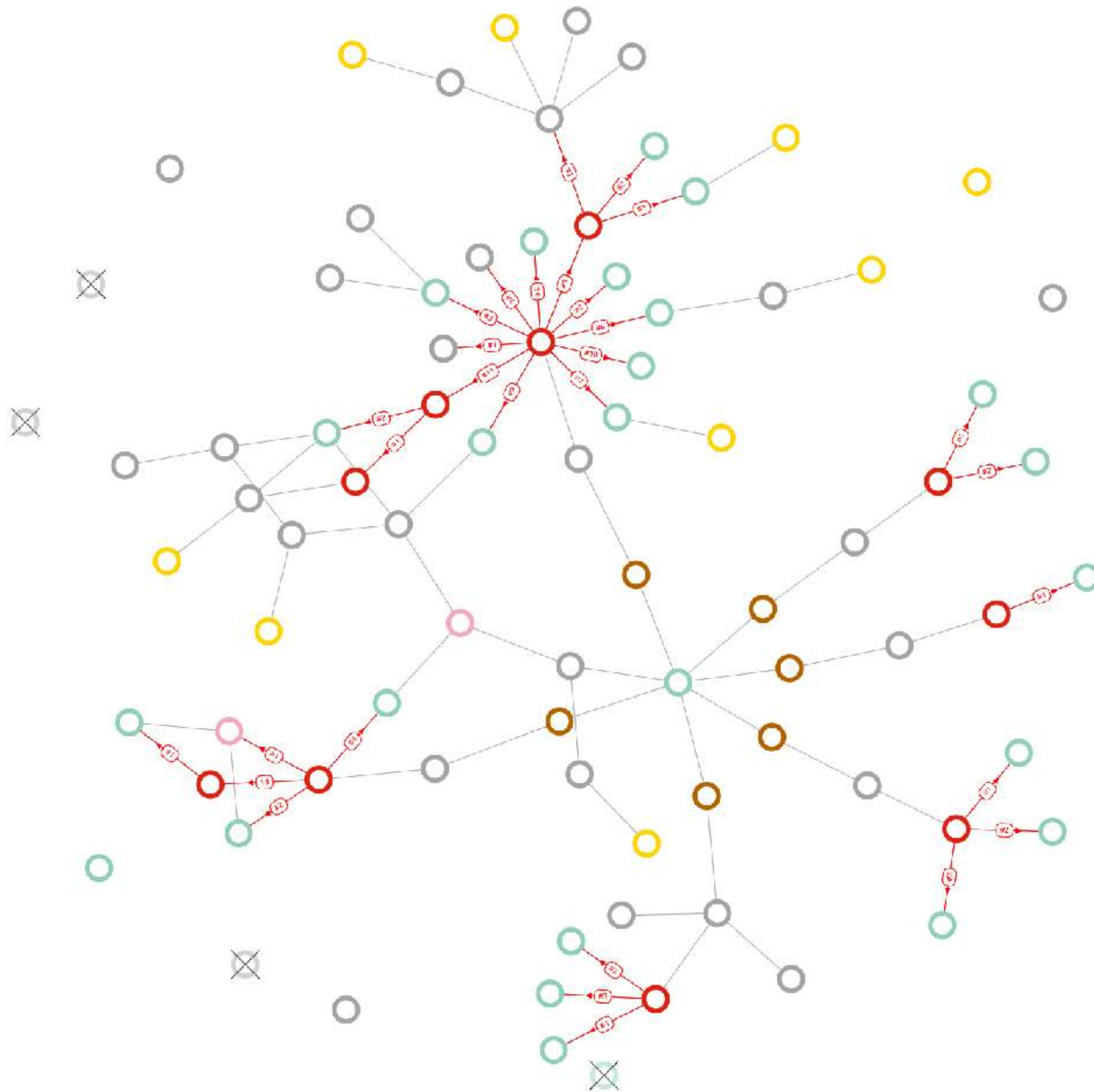
```
{
  "id": "<6847.758.0>",
  "info": {
    "current_function": {
      "arity": 6,
      "function": "loop",
      "module": "gen_server"
    },
    "dictionary": {
      "$ancestors": [
        "<6847.756.0>",
        "Elixir.EveryColor.Endpoint.Server",
        "Elixir.EveryColor.Endpoint",
        "Elixir.EveryColor.Supervisor",
        "<6847.224.0>"
      ],
      "$initial_call": {
        "arity": 1.
      }
    }
  }
}
```

PAPARAZZI

How to spy on actors inside the BEAM

RETHINKING SUPERVISION TREE VISUALISATION

paparazzi



Toggle visualisations

Node labels

Links

Supervision

Message tracing

Restart strategy

Dead processes

Actions

Purge dead

Clear message store

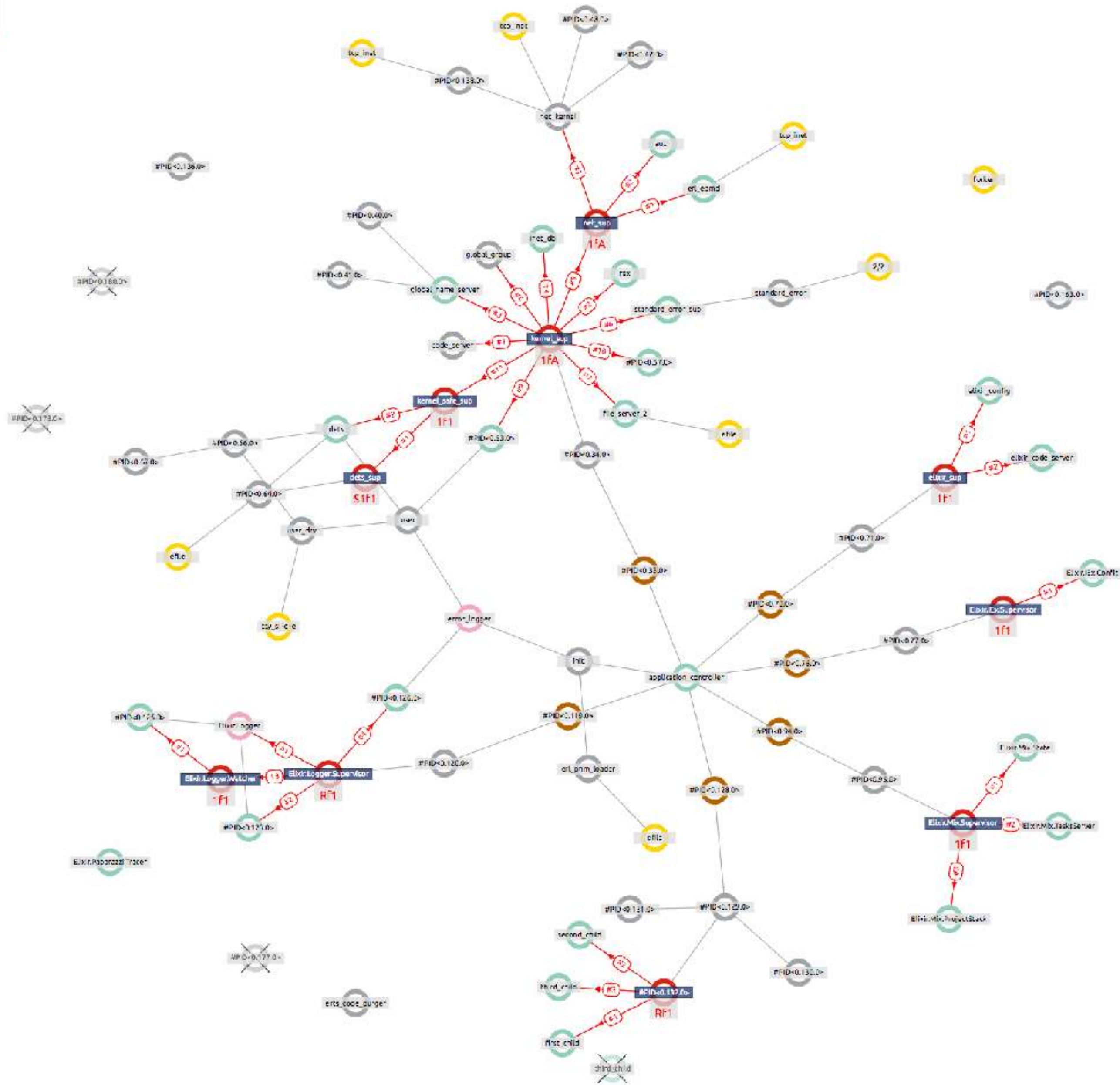
Stop all tracing

Start tracing everything

Pin all nodes

Unpin all nodes

paparazzi



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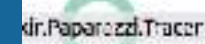
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Unpin all nodes



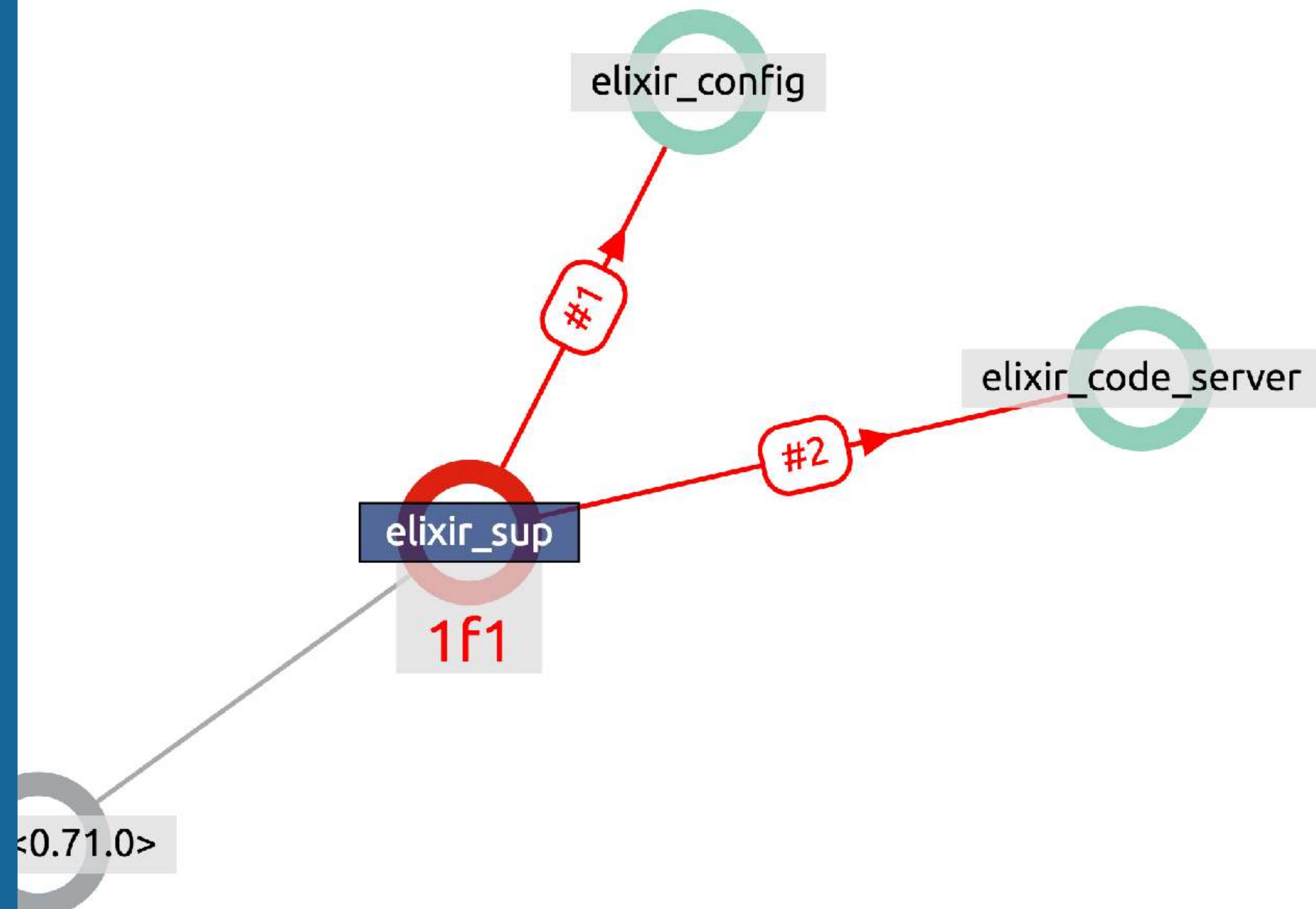
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Pin all nodes

VISUAL SUPERPOSITION

paparazzi



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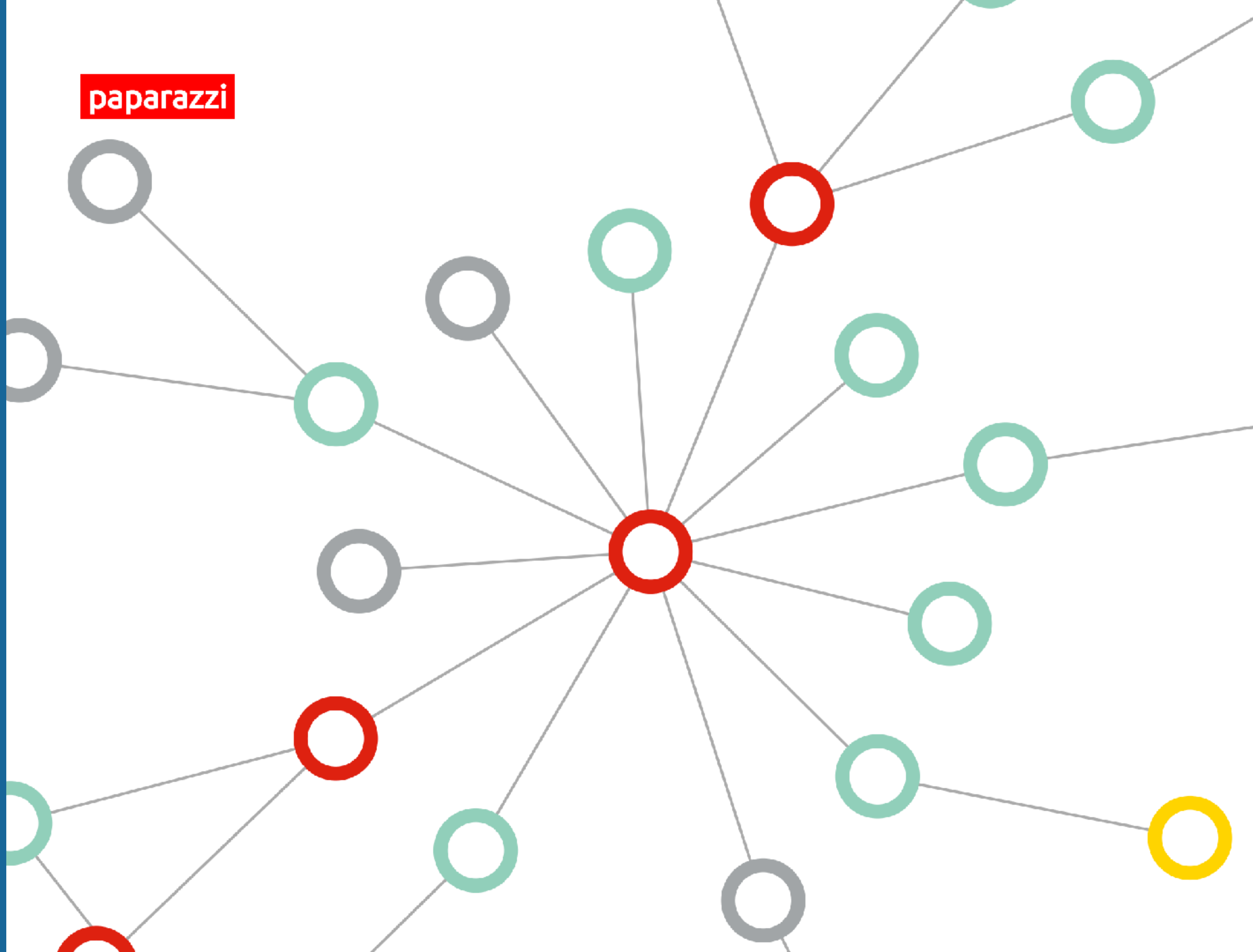
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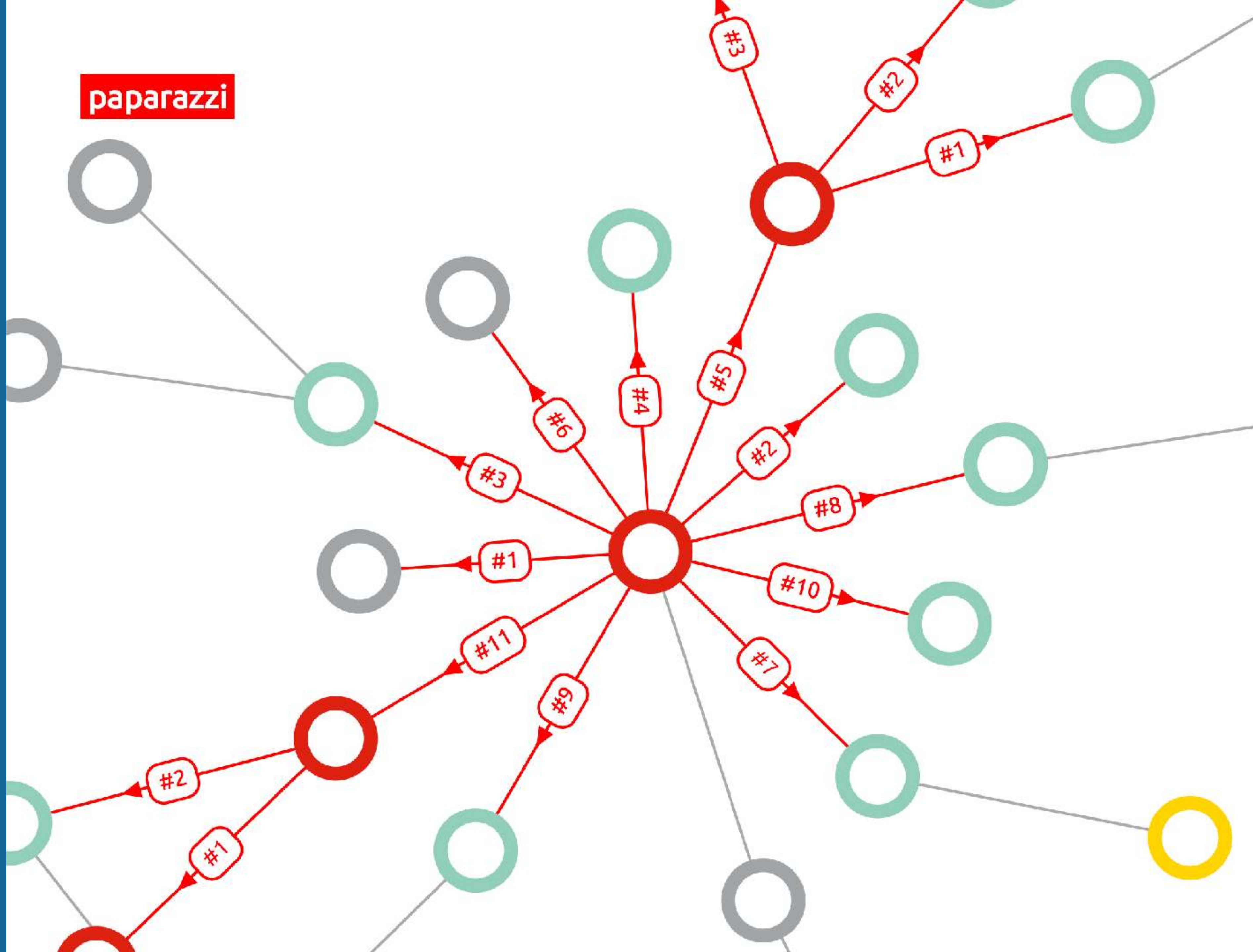
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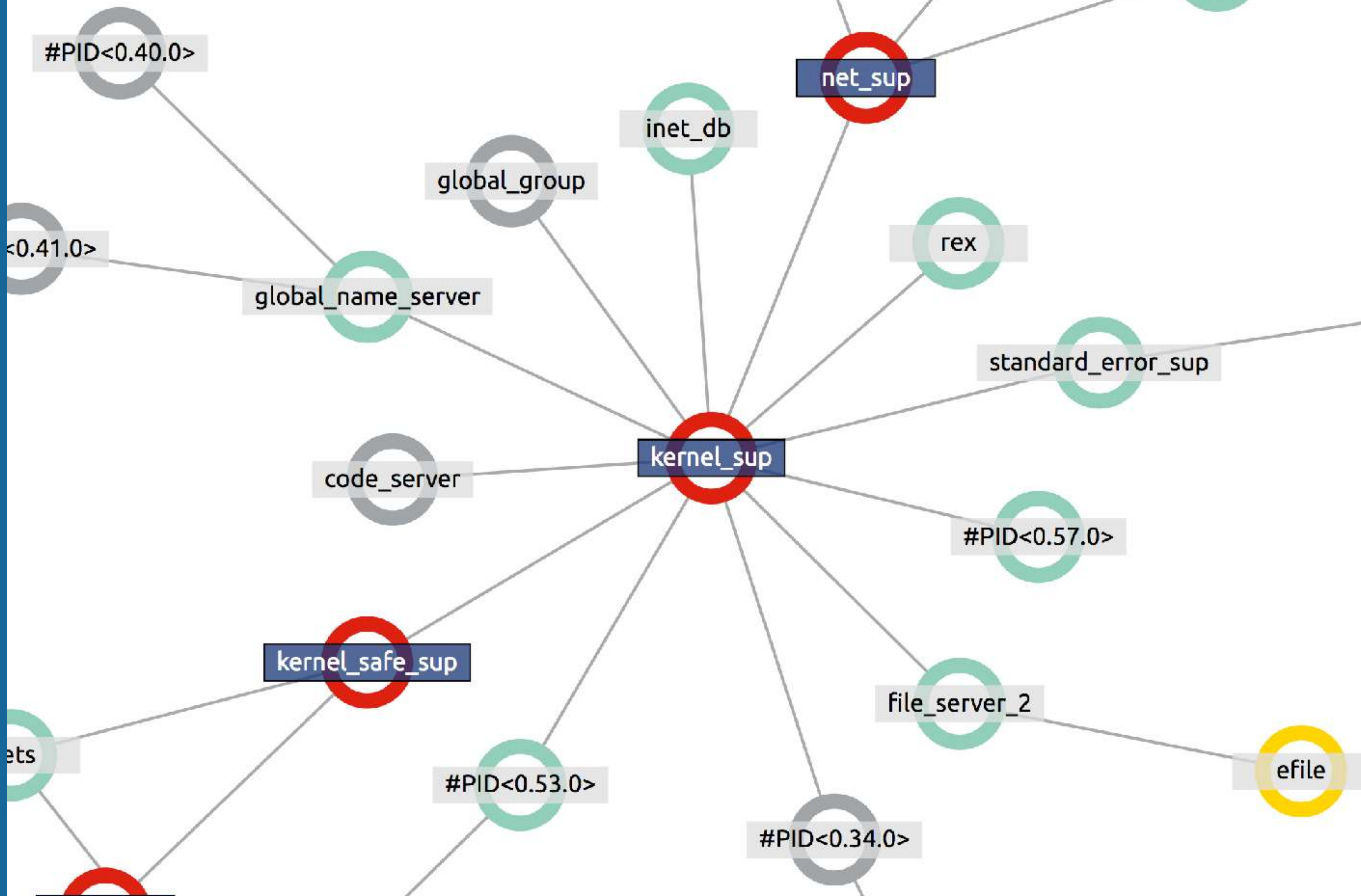
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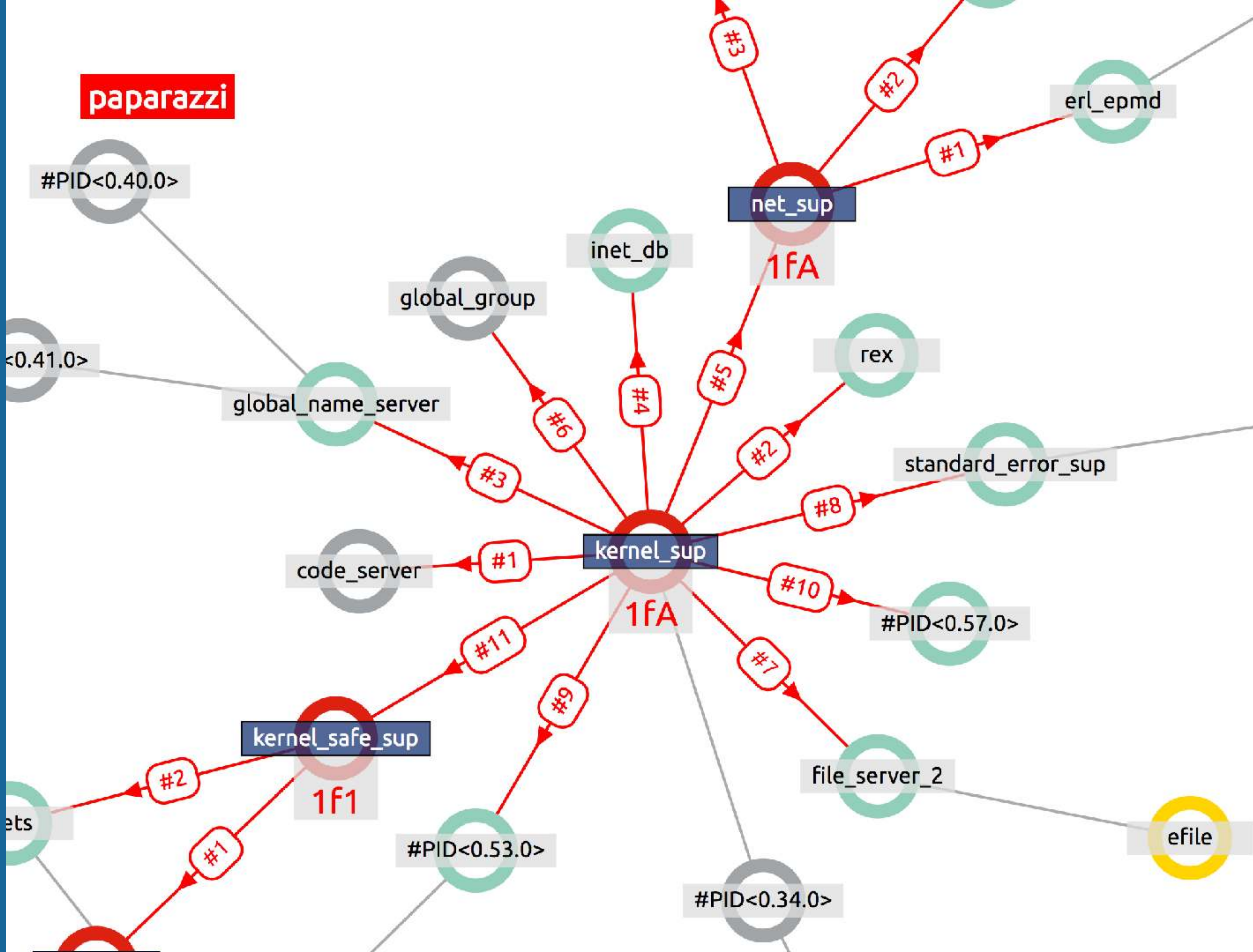
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paparazzi



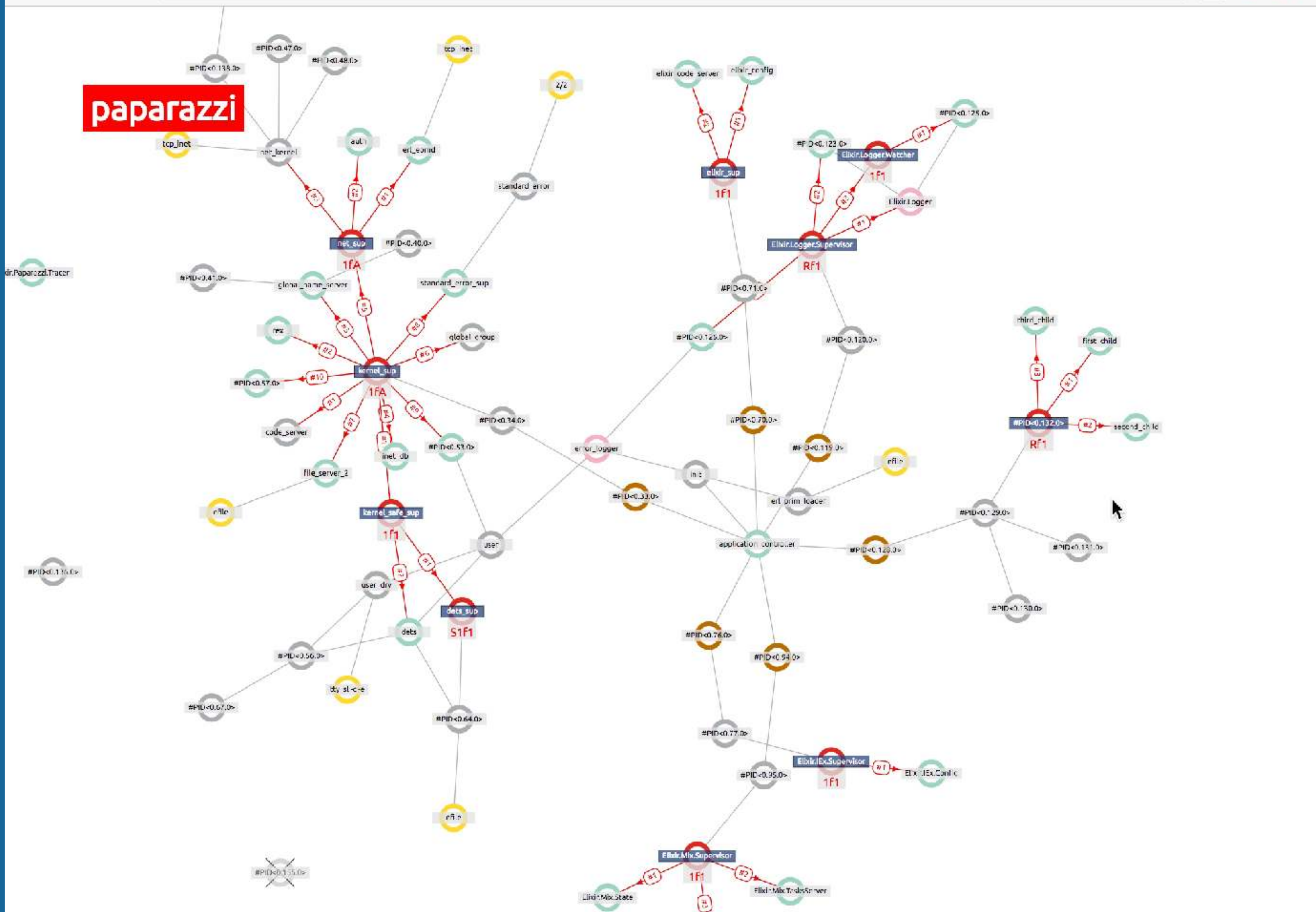
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INTERACTIVE FAULT-TOLERANCY TESTING



Toggle visualisations

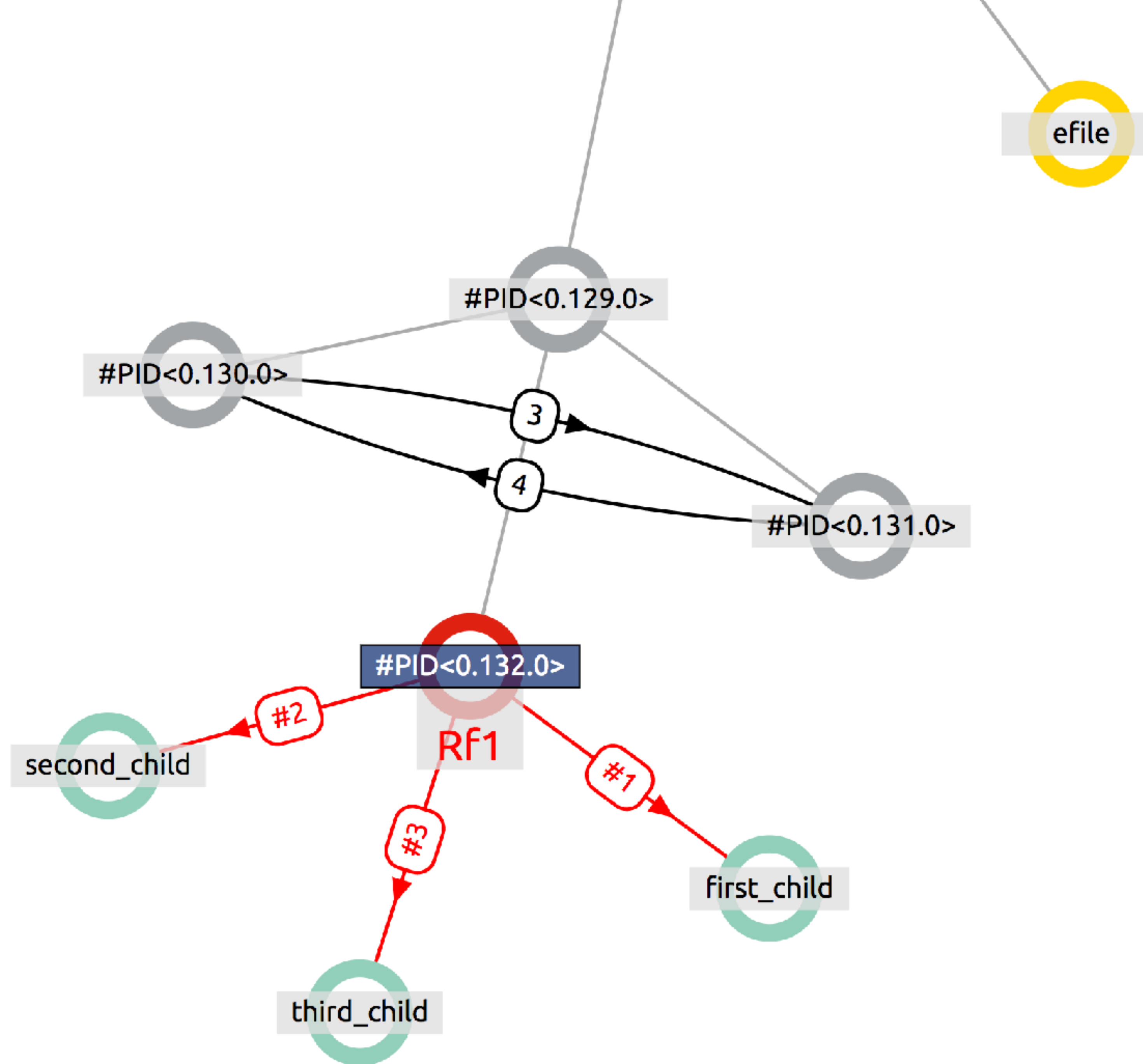
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CONTEXTUAL MESSAGE TRACING

paparazzi



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From: #PID<0.131.0>

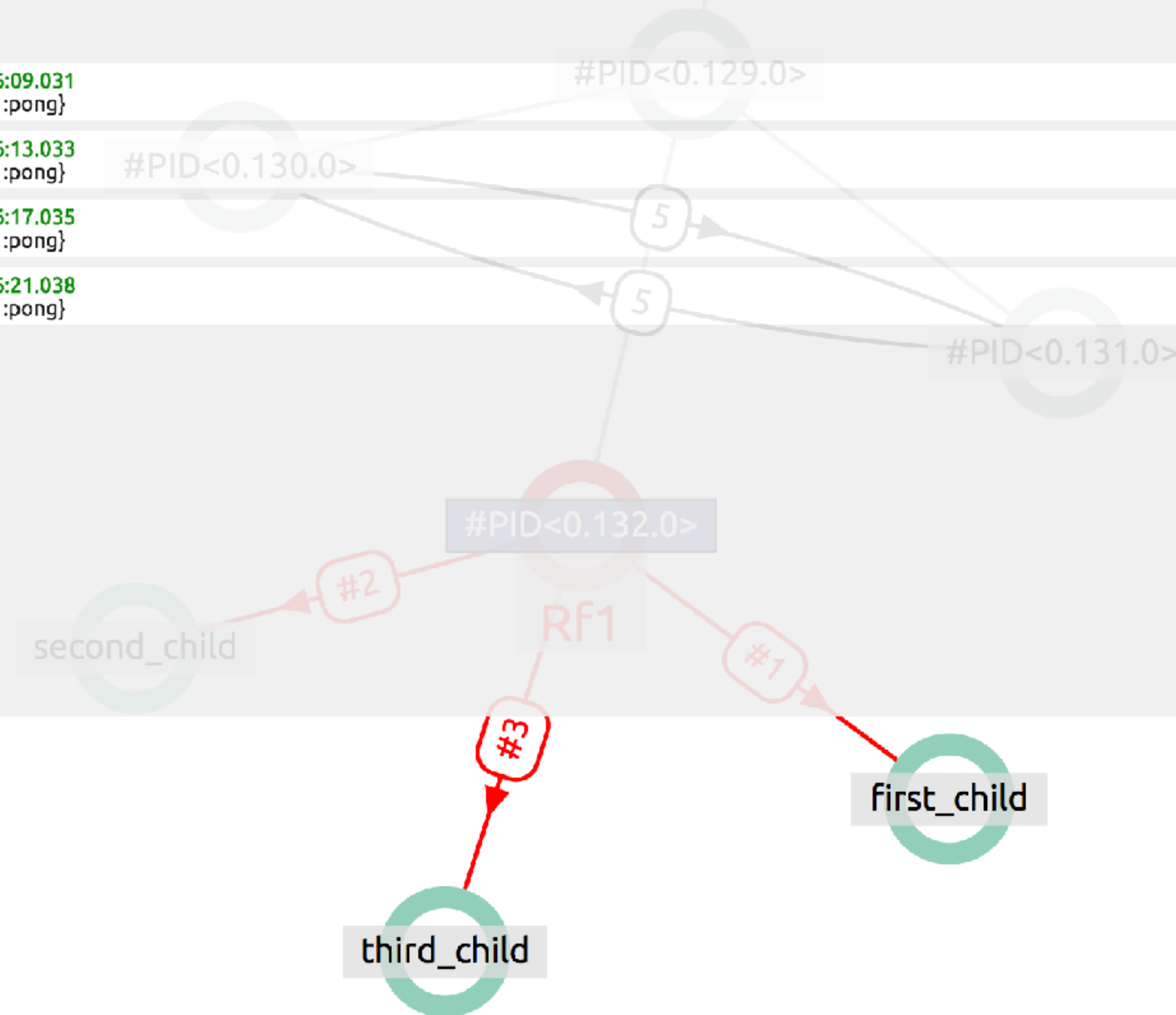
To: #PID<0.130.0>

(#1) 2017.05.04 13:36:09.031
{#PID<17645.131.0>, :pong}

(#2) 2017.05.04 13:36:13.033
{#PID<17645.131.0>, :pong}

(#3) 2017.05.04 13:36:17.035
{#PID<17645.131.0>, :pong}

(#4) 2017.05.04 13:36:21.038
{#PID<17645.131.0>, :pong}



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Unpin all nodes

From: #PID<0.131.0>

To: #PID<0.130.0>

(#1) 2017.05.04 13:36:09.031
{#PID<17645.131.0>, :pong}

(#2) 2017.05.04 13:36:13.033
{#PID<17645.131.0>, :pong}

(#3) 2017.05.04 13:36:17.035
{#PID<17645.131.0>, :pong}

(#4) 2017.05.04 13:36:21.038
{#PID<17645.131.0>, :pong}

#PID<0.129.0>

#PID<0.130.0>

8

9

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rex

#PID<0.53.0>

From: #PID<0.54.0>

To: #Port<0.673>

(#1) 2017.05.04 13:37:17.578

{#PID<17645.54.0>, {:command, [1, 0, 0]}}

(#2) 2017.05.04 13:37:17.579

{#PID<17645.54.0>, {:command, [0 | "\n"]}}

(#3) 2017.05.04 13:37:17.612

{#PID<17645.54.0>, {:command, [5 | "\e[33m\e[35mnil\e[0m\e[33m\e[0m\n"]}}

(#4) 2017.05.04 13:37:17.612

{#PID<17645.54.0>, {:command, [0 | "iex(pingpong@vamosimbpro)3> "]}}

(#5) 2017.05.04 13:37:17.716

{#PID<17645.54.0>, {:command, [1, 0, 0]}}

(#6) 2017.05.04 13:37:17.716

{#PID<17645.54.0>, {:command, [0 | "\n"]}}

(#7) 2017.05.04 13:37:17.719

{#PID<17645.54.0>, {:command, [5 | "\e[33m\e[35mnil\e[0m\e[33m\e[0m\n"]}}

(#8) 2017.05.04 13:37:17.720

{#PID<17645.54.0>, {:command, [0 | "iex(pingpong@vamosimbpro)4> "]}}

(#9) 2017.05.04 13:37:17.854

{#PID<17645.54.0>, {:command, [1, 0, 0]}}

(#10) 2017.05.04 13:37:17.855

{#PID<17645.54.0>, {:command, [1, 0, 0]}}

user_drv

user

tty_sl -c -e

error

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KEY LEARNINGS

**SCALABILITY
EVERYWHERE**

FUTURE PLANS

**[HTTPS://GITHUB.COM/
DIGITALNATIVES/PAPARAZZI](https://github.com/DigitalNatives/Paparazzi)**

THANK YOU! QUESTIONS?

@DVAMOSI

[HTTPS://GITHUB.COM/
DANIELVAMOSI](https://github.com/danielvamosi)